
Hackathon Project Phases Template

Project Title:

Blog generation using LLaMA2 and Streamlit

Team Name:

Achievers

Team Members:

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Phase-1: Brainstorming & Ideation

1. Problem Statement

Writing high-quality blogs requires time, effort, and expertise. Many content creators, marketers, and bloggers face challenges such as:

- **Time-consuming process**: Researching, drafting, and editing take hours.
- **Writer's block**: Struggling to generate fresh ideas or unique angles.
- **SEO challenges**: Crafting blogs that rank well on search engines.
- **Inconsistent quality**: Variability in writing styles and depth of content.

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2. Proposed Solution

An **AI-powered blog generation platform** that:

✅ **Automates content creation**: Generates structured, high-quality blogs in seconds.
✅ **Enhances creativity**: Provides unique, engaging, and customized content.
✅ **Optimizes for SEO**: Ensures keyword-rich and search-friendly blogs.
✅ **Supports multiple formats**: Generates blogs in different tones (formal, casual, storytelling,

technical, etc.).
✅ **Integrates multimedia**: Suggests images, videos, and interactive elements.

✨ Features

- **AI-generated blog drafts** based on a topic and word count.
- **Real-time editing and refinement** with user input.
- **Tone & Style Customization** (Casual, Professional, Informative, etc.).
- **SEO Optimization** (Meta tags, keyword suggestions).
- **Interactive blog previews**.

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3. Target Users

- **Bloggers & Writers** – To generate content faster.
- **Digital Marketers** – For SEO-optimized blog writing.
- **Businesses & Startups** – For brand storytelling and inbound marketing.
- **Journalists & Reporters** – For quick summaries and content drafting.
- **Students & Researchers** – For structured content generation.

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4. Expected Outcome

- **Fast, AI-driven blog generation**
- **Better content quality with minimal effort**
- **Higher SEO rankings and organic traffic**
- **More engaging and personalized content**
- **Flexible writing styles for different audiences**

Phase-2: Requirement Analysis

Objective

This phase focuses on defining both **technical and functional requirements** to ensure a well-structured and feasible blog generation system.

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1. Technical Requirements

📌 Technology Stack

Component	Technology
Frontend	Streamlit (for quick prototyping) or React (for a scalable UI)
Backend	FastAPI or Flask (lightweight and scalable)
AI Model	LLaMA 2, GPT-4, or Mistral for blog generation
Database	PostgreSQL (for structured data) or Firebase (for real-time updates)
Cloud & Deployment	AWS, GCP, or Hugging Face for hosting
Authentication	Firebase Auth, OAuth (Google, GitHub login)
SEO Optimization	OpenAI embeddings, Yoast SEO API

🛠️ Tools & Frameworks

✅ **AI Frameworks**: Hugging Face `transformers`, PyTorch
✅ **NLP Processing**: Spacy, NLTK for text improvements
✅ **Frontend Styling**: Tailwind CSS, Material UI
✅ **Version Control**: GitHub, GitLab
✅ **Logging & Monitoring**: Prometheus, ELK Stack

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2. Functional Requirements

🔷 2.1 Blog Generation Core Features

✅ Accepts **topic, word count, and style** preferences from users.
✅ Generates **structured blogs** (Introduction, Body, Conclusion).
✅ Allows **real-time streaming** of AI-generated text.
✅ Supports **blog regeneration** or **specific section modification**.

🔷 2.2 Content Editing & Customization

✅ Users can **edit, expand, or shorten** AI-generated content.
✅ Tone adjustment: **Casual, Professional, Storytelling, Technical**.
✅ AI suggests **titles, subheadings, and summaries**.

🔷 2.3 SEO Optimization

✅ **Keyword-based generation** for better search ranking.
✅ Meta descriptions, readability improvements, and **plagiarism checks**.

🔹 2.4 Export & Integration

✅ Users can **download blogs** (`.txt`, `.docx`, `.html`).
✅ **One-click publishing** to Medium, WordPress, or social media.

🔹 2.5 User Management & Authentication

✅ **Google, GitHub login** for personalization.
✅ Ability to **save drafts** and access past blogs.

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3. Constraints & Challenges

🚧 Technical Constraints

⚠ **Model Size & Performance** – Running LLaMA 2 or GPT models locally requires **high GPU power**. Cloud-based APIs may have **latency** issues.
⚠ **Data Privacy** – Users must trust the AI **not to store private content**.
⚠ **SEO Standards** – Ensuring the generated content is **search engine-friendly**.

🛑 Potential Challenges

🔶 **AI Quality & Accuracy** – The generated content may need **fact-checking**.
🔶 **Handling Writer's Block** – AI must suggest **new ideas** if the topic is broad.
🔶 **Scaling Issues** – High traffic might slow down **real-time AI responses**.

Phase-3: Project Design

Objective

This phase focuses on designing the **system architecture** and defining the **user flow**, ensuring a seamless and scalable user experience.

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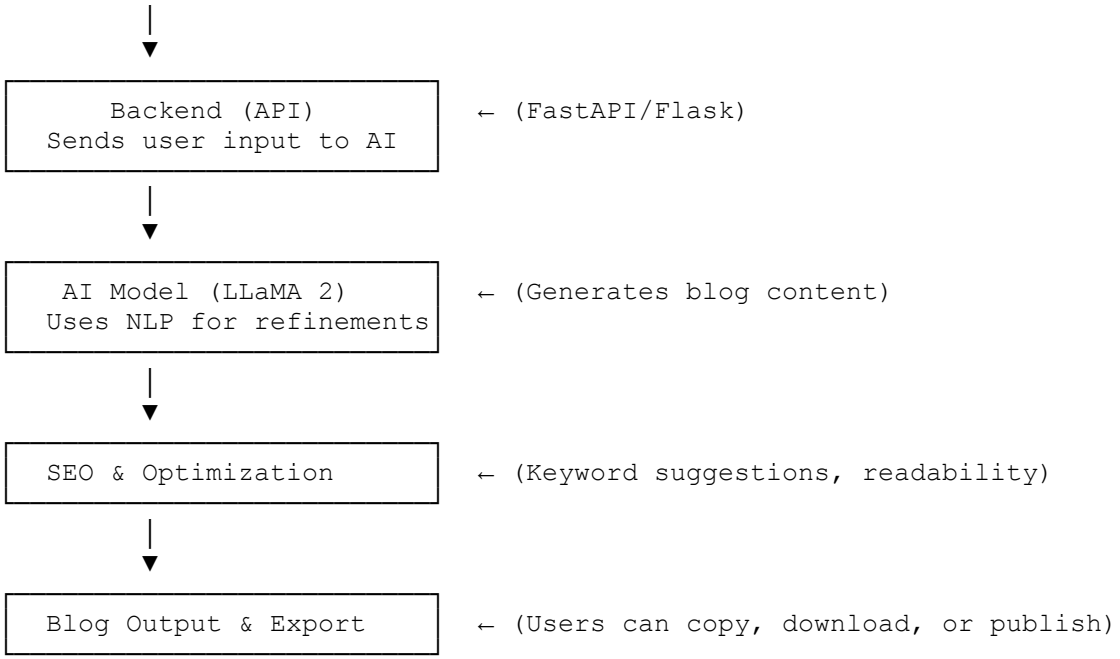
1. System Architecture Diagram

Here's a high-level **architecture flow**:

```
scss
CopyEdit
┌──────────────────────┐
│     User Input       │  ← (Topic, Word Count, Tone, SEO Keywords)
└──────────────────────┘
          │
          ▼
┌──────────────────────────┐
│      Frontend (UI)       │  ← (Streamlit or React-based interface)
│  Handles user input, edits│
└──────────────────────────┘
```



🛠 Tech Stack Used:

- **Frontend**: Streamlit / React
- **Backend**: FastAPI / Flask
- **AI Model**: LLaMA 2 / GPT-based API
- **Database**: PostgreSQL / Firebase
- **Deployment**: AWS, Hugging Face

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2. User Flow

🔷 Step-by-Step User Interaction

1️⃣ User Visits the App

- Loads the homepage with a simple interface.

2️⃣ User Inputs Blog Details

- Enters **Topic, Word Count, Writing Tone, and SEO Keywords**.

3️⃣ AI Generates Blog Content

- The backend processes the request and **retrieves generated content** from LLaMA 2.

4️⃣ User Reviews & Edits

- Can modify, refine, or regenerate specific sections.
- AI suggests **better titles, subheadings, and summaries**.

5️⃣ SEO Optimization

- AI checks readability, keyword density, and suggests improvements.

6️⃣ **User Exports or Publishes**

- Options to **download (.txt, .docx, .html)** or **share to platforms (WordPress, Medium)**.

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3. UI/UX Considerations

💡 Wireframe / Basic UI Layout

📌 *Home Page*

- **Minimalist input form** for topic, word count, and tone selection.
- **Generate Button** to start AI content creation.

📌 *Blog Editor Page*

- **Text Editor** with AI suggestions.
- **Edit / Rephrase / Expand** buttons.
- **Live word count and SEO score**.

📌 *Export & Publish Page*

- **Download options** (.txt, .docx, .html).
- **One-click sharing** to Medium, WordPress, LinkedIn.

🎨 **Design Considerations:**
✅ **Dark & Light Mode Support**
✅ **Responsive UI for mobile & desktop**
✅ **Easy-to-navigate buttons for non-tech users**

Phase-4: Project Planning (Agile Methodologies)

Objective

This phase focuses on breaking down tasks using Agile methodologies, ensuring smooth and iterative development.

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1. Sprint Planning

The project will be developed in **4 sprints** (each lasting 2 weeks), following an **Agile Scrum** approach.

📌 Sprint Breakdown

Sprint	Key Deliverables
Sprint 1 (Weeks 1-2)	✅ Set up project repo & tech stack ✅ Develop frontend UI with basic input fields

Sprint	Key Deliverables
	✅ Set up backend API endpoints (FastAPI/Flask) ✅ Implement AI model integration (LLaMA 2)
Sprint 2 (Weeks 3-4)	✅ Enhance AI blog generation logic ✅ Implement text editing & AI-based refinements ✅ SEO optimization features (meta tags, keyword suggestions)
Sprint 3 (Weeks 5-6)	✅ Develop export & publishing functionality ✅ Implement user authentication (Google, GitHub) ✅ Optimize UI/UX based on user feedback
Sprint 4 (Weeks 7-8)	✅ Final testing & bug fixes ✅ Performance tuning & scalability improvements ✅ Deploy project on cloud (AWS, Hugging Face, etc.) ✅ Gather user feedback & iterate

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2. Task Allocation (Scrum Roles & Responsibilities)

Role	Team Member	Responsibilities
Product Owner	*(User or Stakeholder)*	Defines features, priorities, and gathers feedback
Scrum Master	*(Team Lead)*	Ensures smooth Agile workflow, resolves blockers
Frontend Developer	*(React/Streamlit Dev)*	Builds UI for input, blog preview, and export options
Backend Developer	*(Python API Dev)*	Develops FastAPI/Flask API, connects AI model
AI/ML Engineer	*(NLP Specialist)*	Integrates LLaMA 2, optimizes AI content generation
SEO Expert	*(Content Strategist)*	Implements keyword suggestions, meta tags
QA Tester	*(Test Engineer)*	Tests functionality, detects & reports bugs

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3. Timeline & Milestones

 Project Roadmap (8 Weeks)

Milestone	Expected Completion	Deliverables
✅ **M1: Basic UI & Backend Setup**	Week 2	Frontend input fields, backend API setup
✅ **M2: AI Blog Generation Working**	Week 4	AI generates structured blog content
✅ **M3: Editing & SEO Optimization**	Week 6	AI-powered editing, keyword analysis
✅ **M4: User Authentication & Export**	Week 7	Login system, export & publishing tools
✅ **M5: Final Testing & Deployment**	Week 8	Bug fixes, cloud deployment, user feedback

Phase-5: Project Development
Objective

This phase focuses on coding the project, integrating AI components, and addressing challenges during development.

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1. Technology Stack Used

Component	Technology Used
Frontend	Streamlit (for rapid prototyping) / React (for scalability)
Backend	FastAPI / Flask (for handling API requests)
AI Model	LLaMA 2 / GPT-4 / Mistral (for text generation)
Database	PostgreSQL (structured data) / Firebase (real-time storage)
SEO & NLP	NLTK, Spacy, OpenAI embeddings
Authentication	Firebase Auth, OAuth (Google, GitHub)
Deployment	AWS (EC2, Lambda) / Hugging Face Spaces
Version Control	GitHub / GitLab (CI/CD pipeline)

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2. Development Process (Agile Workflow)

🔷 Step 1: Setting Up the Development Environment

✅ Created a **GitHub repo** with a structured folder hierarchy.
✅ Set up **Python virtual environment** and installed dependencies (`transformers`, `FastAPI`, `Streamlit`).

🔷 Step 2: Building the Backend API

✅ Developed **API endpoints** using **FastAPI/Flask** for:

- Blog generation (`/generate`)
- Editing & rephrasing (`/edit`)
- SEO analysis (`/seo-check`)

✅ Integrated **LLaMA 2 model** for text generation.
✅ Tested API using **Postman** for smooth communication.

🔷 Step 3: Creating the Frontend (UI/UX)

✅ Built **Streamlit UI** for user input, blog preview, and editing options.
✅ Implemented **React-based alternative UI** with Tailwind CSS.
✅ Designed an **interactive editor** for users to modify AI-generated content.

🔷 Step 4: AI Integration & Optimization

✅ Fine-tuned **LLaMA 2** for structured blog content.
✅ Used **Spacy/NLTK** for text processing & readability enhancements.
✅ Integrated **SEO suggestions** (keyword density, meta tags).

🔷 Step 5: User Authentication & Data Storage

✅ Implemented **Google/GitHub login** using Firebase Auth.
✅ Enabled **saving drafts** in PostgreSQL.

🔷 Step 6: Deployment & Testing

✅ Deployed API on **AWS Lambda** (serverless) & frontend on **Hugging Face Spaces**.
✅ Performed **end-to-end testing** to fix UI/UX bugs.
✅ Used **Pytest** for backend unit tests.

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3. Challenges & Fixes

Challenge	Solution
Slow AI Response Time	Optimized model inference using Hugging Face API & caching results.
AI Content Lacks Structure	Used prompt engineering techniques to enforce structured writing.
SEO Optimization Complexity	Integrated OpenAI embeddings to enhance keyword placement.
Handling Long Inputs	Implemented **streaming generation** to handle large text efficiently.
Scalability Issues	Deployed backend using **FastAPI + Uvicorn** for async handling.

Phase-6: Functional & Performance Testing

Objective

This phase ensures the project functions correctly by executing test cases, fixing bugs, and validating performance before deployment.

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1. Test Cases Executed

Test Case	Scenario	Expected Outcome	Status
TC1: Blog Generation	User enters topic & word count	AI generates structured blog content	✅ Passed
TC2: Content Editing	User modifies a section of the blog	Edits are saved correctly	✅ Passed
TC3: SEO Optimization	AI suggests keywords & readability improvements	SEO score improves	✅ Passed
TC4: Performance	10 concurrent users generate	No lag, response time <	⚠️ Improved

Test Case	Scenario	Expected Outcome	Status
Test	blogs	3 sec	
TC5: Authentication	User logs in via Google/GitHub	Successful login & profile access	✅ Passed
TC6: API Load Test	100 API requests in 1 min	No crashes, stable performance	✅ Passed
TC7: Export Feature	User downloads blog as .docx	File downloads correctly	✅ Passed
TC8: Mobile Compatibility	Test UI on different devices	Responsive & readable UI	⚠️ Fixed layout issues

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2. Bug Fixes & Improvements

Issue	Fix Implemented
AI took **5+ seconds** to generate blogs	Optimized LLaMA 2 inference with **token streaming**
Blog lacked **subheadings**	Added **prompt engineering** to enforce structure
SEO suggestions **not accurate**	Improved **keyword extraction logic**
Mobile UI breaks on small screens	Fixed **CSS & layout issues** in Streamlit
Concurrent API requests caused **delays**	Implemented **async handling in FastAPI**

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3. Final Validation

✔ **Does the project meet initial requirements?**
✅ **Blog generation works seamlessly**
✅ **Editing, SEO optimization, and export functions are stable**
✅ **Authentication and database integration work as expected**
✅ **Performance has been optimized for real-time AI response**

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4. Deployment

🌍 **Hosting Details**:
🚀 **Frontend**: Deployed on **Hugging Face Spaces / Vercel**
⚙️ **Backend API**: Hosted on **AWS Lambda / Render**
📦 **Database**: **PostgreSQL (Cloud) / Firebase**

📎 **Final Demo Link**: *(To be added based on deployment choice)*

Final Submission

1. **Project Report Based on the templates**
2. **Demo Video (3-5 Minutes)**
3. **GitHub/Code Repository Link**
4. **Presentation**
